Biostatistics 1 - Assignment 1

Group 3: William Fleury Dziuk, Galya Gadzhalova, Chen Ma, Afroditi-Iliana Zotou

2025-12-08

Introduction

Here we study the outcome coronary heart disease (CHD, variable name in our data:chd69). Our task is to find a prediction model for the risk of CHD. We will consider the following covariates/predictors which are previously known risk factors for heart disease: behaviour type A/B, age, systolic bp, cholesterol, bmi, smoking, and arcus.

We have access to the dataset wcgs or the Western Collaborative Group Study data. This is a prospective cohort study, that recruited middle-aged men (ages 39 to 59) who were employees of 10 California companies and collected data on 3154 individuals during the years 1960-1961. These subjects were primarily selected to study the relationship between behavior pattern and the risk of coronary hearth disease (CHD). A number of other risk factors were also measured to provide the best possible assessment of the CHD risk associated with behavior type. Additional variables collected include age, height, weight, systolic blood pressure, diastolic blood pressure, cholesterol, smoking, and corneal arcus

Preparing for analysis and cleaning the data

In this analysis we'll consider the following variables:

chd69 age systolic bp cholesterol bmi smoking arcus behaviour type

A new dataframe was created containing only the variables relevant for the analysis.

```r
d <- wcgs %>% select(id,agegroup,age0,cholmmol,sbp10,bmi,smokerf,arcus0,dibpat0f,chd69)
```

After that, all complete cases were extracted from the cleaned dataset.

```r
# Complete cases of the cleaned data*
dc <- d %>% drop_na()
```

Task 1 - Table 1

Task: Create table 1. Describe all available variables in the data. We will continue with the complete case data we have now called dc

Answer:: The following table summarizes the characteristics of the complete case dataset (dc) for the variables we are considering.

Continuous variables are presented as mean (Standard Deviation - SD) and categorical variables are presented as number (n) (%). Our complete case data (dc) includes $n = 3139$ individuals.

```
                      Overall
n                        3139
id (mean (SD))        10480.53 (5882.89)
agegroup (%)
   [39,45)               1440 (45.9)
   [45,55)               1380 (44.0)
   [55,60]                319 (10.2)
age0 (mean (SD))        46.28 (5.52)
cholmmol (mean (SD))     5.80 (1.10)
sbp10 (mean (SD))       12.86 (1.51)
bmi (mean (SD))         24.52 (2.56)
smokerf = Yes (%)        1494 (47.6)
arcus0 (mean (SD))       0.30 (0.46)
dibpat0f = A (%)         1582 (50.4)
chd69 (mean (SD))        0.08 (0.27)
```

Task 2 - Overall risk or overall rate

Task: What is the outcome we are interested in?

Answer: We are interested in the outcome coronary heart disease (CHD). In our dataset the variable for this is chd69. Overall, our task is to find a prediction model for the risk of CHD.

Task: What are the known risk factors for our outcome of interest?

Answer: The known risk factors (covariates/predictors) for our outcome of interest, Coronary Heart Disease (CHD), are: Behaviour type (Type A/B), Age, Systolic blood pressure (systolic bp), Cholesterol, BMI, Smoking, Arcus (corneal arcus)

These are the variables that have been previously known as risk factors for heart disease and are taken into consideration for our model.

The known factors that cause risk for the particular disease are separated into two groups (modifiable and non-modifiable) [@cdc_heart_disease_risk_factors].

Modifiable risk factors are:

- *smoking*
- *physical inactivity*
- *unhealthy diet*
- *high blood pressure*
- *unhealthy blood cholesterol levels*

Non-modifiable risk factors are:

- *sex*
- *ethnicity*
- *age*
- *family history (genetics)*

Task: How many people are included?

Answer: The dataset used for this analysis contains 3.139 observations.

Task: What is the overall prevalence of the disease in our cohort?

Answer: To find the overall prevalence (the percentage of people who have the disease), we look at the mean value of the binary outcome variable (chd69) reported in Table 1. So, the overall prevalence is 8%

Task 3 - Building the model and choosing predictors

3a. Task: Use the available data to create the optimal prediction model for our outcome, choose the appropriate model and the predictors that will improve your predictions. Please explain the reasons for choosing the model type you chose and why the predictors were chosen.

Answer:

Before building a predictive model, a correlation matrix was calculated to check for variables with strong correlation. The threshold for strong correlation is 0.8.

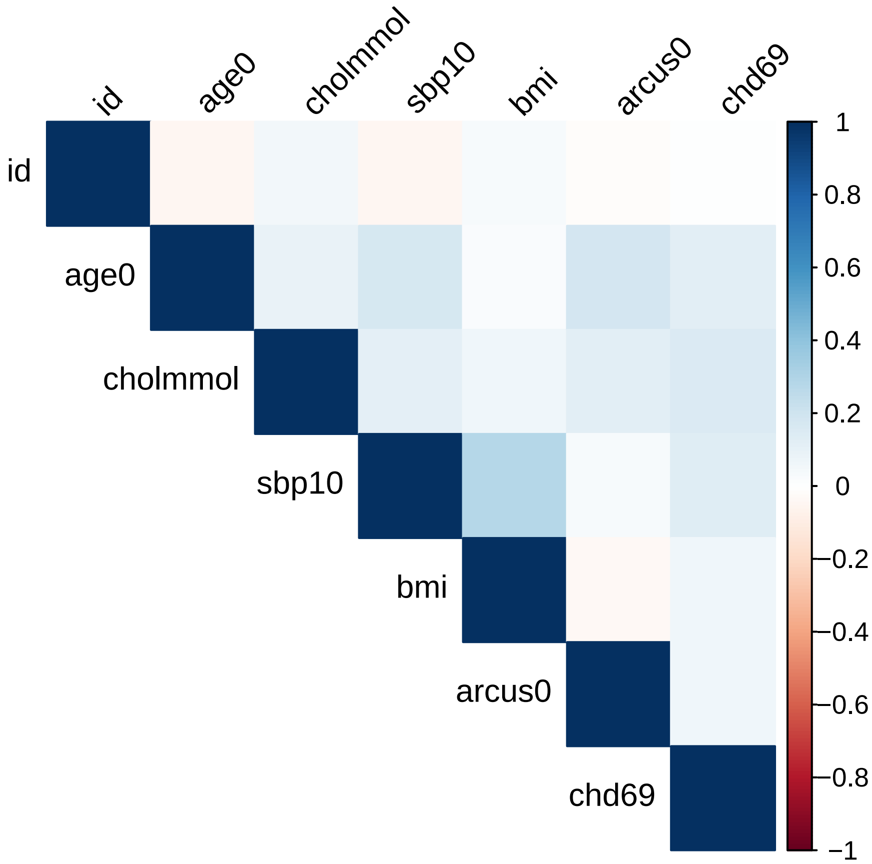


Table 1: Correlation Matrix of Numeric Variables

	id	age0	cholmmol	sbp10	bmi	arcus0	chd69
id	1.00	-0.05	0.06	-0.05	0.03	-0.02	0.00
age0	-0.05	1.00	0.09	0.17	0.03	0.19	0.12
cholmmol	0.06	0.09	1.00	0.12	0.07	0.13	0.15
sbp10	-0.05	0.17	0.12	1.00	0.29	0.03	0.13
bmi	0.03	0.03	0.07	0.29	1.00	-0.03	0.06
arcus0	-0.02	0.19	0.13	0.03	-0.03	0.19	0.07
chd69	0.00	0.12	0.15	0.13	0.06	0.07	1.00

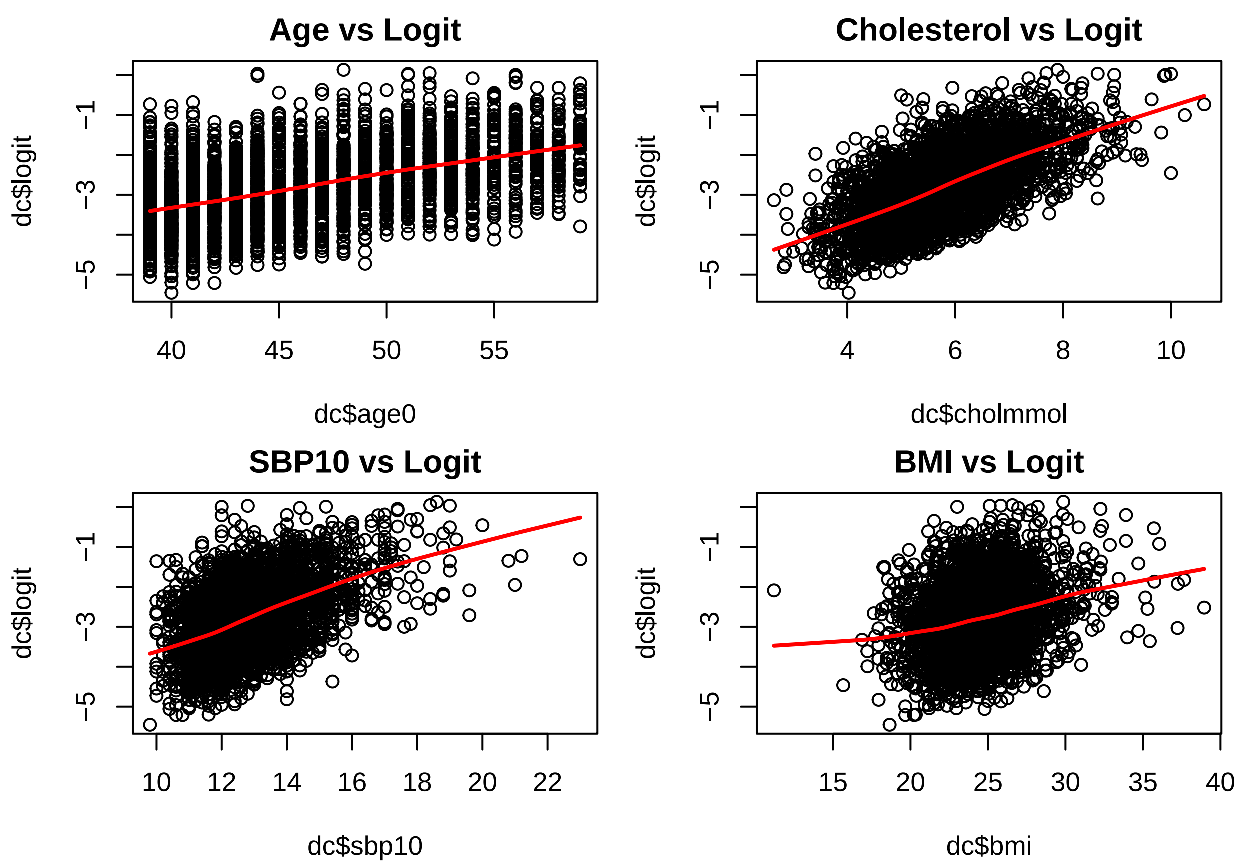
The matrix shows that none of the variables exceed the threshold of 0.8, therefore there is no risk for multicollinearity in this dataset. Meaning that it is not necessary to remove any of the variables at this stage of the analysis, since every variable contributes uniquely.

A logistic regression model was selected for this analysis because the outcome variable is binary and diagnostic checks indicated no evidence of multicollinearity among the predictors.

The next step was to fit the full logistic regression model which included all available predictors in the cleaned dataset. This model was then compared to a reduced model obtained using backward stepwise selection, where predictors were removed based on their contribution to the model fit. In our case the function only excluded one of the initial variables (agegroup).

The result yelds that the step model had better performance compared to the original one. This is concluded since its AIC value is lower (namely 1594.23) than in the full model (namely AIC of 1597.07). This result is also conceptually reasonable since agegroup and age0 represent the same underlying information and including both of them in the model introduces redundancy. Moreover, age0 is more informative as a continuous variable and agegroup was appropriately removed by the stepwise algorithm.

To assess whether the continuous predictors satisfy the logistic regression assumption of linearity in the logit, scatterplots of each predictor against the predicted log-odds were created. They show that the requirement for linearity is met and step-wise logistic regression could be used.



In the following table the coefficients of the step-wise model are presented. Only one of them is not statistically significant.

Table 2: Logistic Regression Coefficient Table

Variable	Estimate	Std_Error	Z_value	P_value
(Intercept)	-12.2494	0.9865	-12.417	< 2e-16
age0	0.0573	0.0123	4.669	3.03e-06
cholmmol	0.4014	0.0600	6.686	2.30e-11
sbp10	0.1820	0.0416	4.380	1.18e-05
bmi	0.0593	0.0266	2.229	0.0258
smokerfYes	0.6002	0.1419	4.230	2.34e-05

Variable	Estimate	Std_Error	Z_value	P_value
arcus0	0.2204	0.1434	1.537	0.1243
dibpat0fA	0.7014	0.1451	4.834	1.34e-06

Therefore, the chosen step-wise model has the following form:

$$\begin{aligned}\operatorname{logit}\big(P(\mathrm{CHD}_i = 1)\big) &= \beta_0 + \beta_1 \cdot \mathrm{age0}_i + \beta_2 \cdot \mathrm{cholmmol}_i + \beta_3 \cdot \mathrm{sbp10}_i \\ &\quad + \beta_4 \cdot \mathrm{bmi}_i + \beta_5 \cdot \mathrm{smoker}_i + \beta_6 \cdot \mathrm{arcus0}_i + \beta_7 \cdot \mathrm{dibpat0fA}_i.\end{aligned}$$

Below are presented the AIC (Akaike Information Criterion) and BIC (Bayesian Information Criterion) values of the full model and the stepwise model. In general, the lower the AIC and BIC values, the more optimal the model is. Both criteria favored the stepwise model, which indicates that the stepwise model provides a better balance between goodness of fit and model complexity.

Table: Comparison of AIC and BIC Between Models

Metric	Value
AIC (Full Model)	1597.072
AIC (Stepwise Model)	1594.227
BIC (Full Model)	1657.589
BIC (Stepwise Model)	1642.640

3b. Task: Are there any predictors that should be included as interaction terms with other variables or as categorical variables? If so, please explain the reasoning for this in the model?

Answer:

Several interaction terms were tested to assess whether the effect of one predictor on CHD depended on another predictor. We mainly investigated two-way interactions, since three or four-way interactions would be difficult to interpret clinically and would potentially lead to overfitting.

The interactions were selected based on clinical reasoning and included age × cholesterol, age × systolic blood pressure, smoking × cholesterol, smoking × systolic blood pressure, and BMI × cholesterol.

First, a model containing all interaction terms was fitted. Then, individual models were fitted in which each interaction term was added separately together with all main predictors. For each model, AIC values were compared with those from the stepwise model and the significance of the interaction term was evaluated. Across these models, most interaction terms did not improve model performance and were not statistically significant.

Only the BMI × cholesterol interaction improved the model by lowering the AIC and showing statistical significance.This model achieved the lowest AIC (1588.306) and the lowest BIC (1642.771). This indicates that, while most predictors act independently in this dataset, the BMI × cholesterol interaction slightly modifies the effect of one predictor on CHD. The above can be verified from the following table.

```
                             AIC      BIC
1. Stepwise Model (Base) 1594.227 1642.640
2. Age x Cholesterol     1594.258 1648.723
3. Age x SBP             1594.788 1649.253
4. Smoker x Cholesterol  1594.314 1648.779
5. Smoker x SBP          1595.996 1650.461
6. BMI x Cholesterol     1588.306 1642.771
```

Furthermore, a likelihood ratio test was conducted to compare the stepwise-selected main-effects model with the model that included the BMI × cholesterol interaction term.

Table 4: Likelihood Ratio Test Comparing Models With and Without the BMI × Cholesterol Interaction

Model	Residual_Df	Residual_Deviance	Df	Deviance	P_value
Stepwise Model (no interaction)	3131	1578.2			
Interaction Model (with bmi:cholmmol)	3130	1570.3	1	7.9208	0.004887

The p-value in the table represents the deviance difference. A high p-value means smaller chi-square statistic and hence smaller deviance difference ΔD. In our case, the p-value = $0.0049 < 0.01$ indicates that this reduction is statistically significant. This result suggests that the effect of cholesterol on CHD risk depends on BMI, and that including the bmi × cholmmol interaction provides additional information and significantly improves the model fit.

After the aforementioned tests, the final model that is chosen is:

$$\text{CHD} \sim \text{Age}+\text{Cholesterol}+\text{Systolic BP}+\text{BMI}+\text{Smoking Status}+\text{Corneal Arcus}+\text{Behavior Type}+\text{Cholesterol} : \text{BMI}$$

and is mathematically expressed as:

$$\begin{aligned}\operatorname{logit}\big(P(\text{CHD}_i = 1)\big) &= \beta_0 + \beta_1 \cdot \text{age0}_i + \beta_2 \cdot \text{cholmmol}_i + \beta_3 \cdot \text{sbp10}_i \\ &\quad + \beta_4 \cdot \text{bmi}_i + \beta_5 \cdot \text{smoker}_i + \beta_6 \cdot \text{arcus0}_i \\ &\quad + \beta_7 \cdot \text{dibpat0fA}_i + \beta_8 \cdot (\text{cholmmol}_i \cdot \text{bmi}_i).\end{aligned}$$

Table 5: Logistic Regression Coefficients for the Model Including the BMI × Cholesterol Interaction

Variable	Estimate	Std_Error	Z_value	P_value
(Intercept)	-21.2833	3.3053	-6.439	1.20e-10
age0	0.0564	0.0123	4.595	4.32e-06
cholmmol	1.8755	0.5187	3.616	0.00030
sbp10	0.1868	0.0416	4.492	7.05e-06
bmi	0.4195	0.1277	3.284	0.00102
smokerfYes	0.5978	0.1420	4.211	2.55e-05
arcus0	0.2190	0.1438	1.523	0.12771
dibpat0fA	0.7266	0.1456	4.989	6.07e-07
cholmmol:bmi	-0.0591	0.0206	-2.864	0.00418

From Table 5, where the numerical estimates, errors, and significance tests for each β are presented, we can also verify that the interaction term (cholmmol:bmi) is statistically significant.

3c. Task: Calculate the predicted risk of the outcome for every person in the dataset according to the model you have chosen and add it to your dataset.

Answer:

We calculated the predicted risk of CHD for each individual in the dataset using the chosen interaction model (*bmi × cholmmol*) and stored these values in a new column called predicted_risk. The table below shows the top rows of the final dataset:

```r
dc$predicted_risk <- predict(interaction_model_bmi_cholmmol, newdata = dc, type = "response")
kable(head(dc), digits=3)
```

id	agegroup	age0	cholmmol	sbp10	bmi	smokerf	arcus0	dibpat0f	chd69	logit	predicted_risk
2001	[45,55)	49	5.769	11.0	19.790	Yes	0	A	0	-2.651	0.060
2002	[39,45)	42	4.538	15.4	22.957	Yes	1	A	0	-2.337	0.076
2003	[39,45)	42	4.641	11.0	23.628	No	0	B	0	-4.579	0.009
2004	[39,45)	41	3.385	12.4	23.111	Yes	0	B	0	-4.316	0.010
2005	[55,60]	59	6.538	14.4	21.523	Yes	1	B	1	-1.529	0.187
2006	[39,45)	44	4.667	15.0	27.667	No	0	B	0	-3.486	0.037

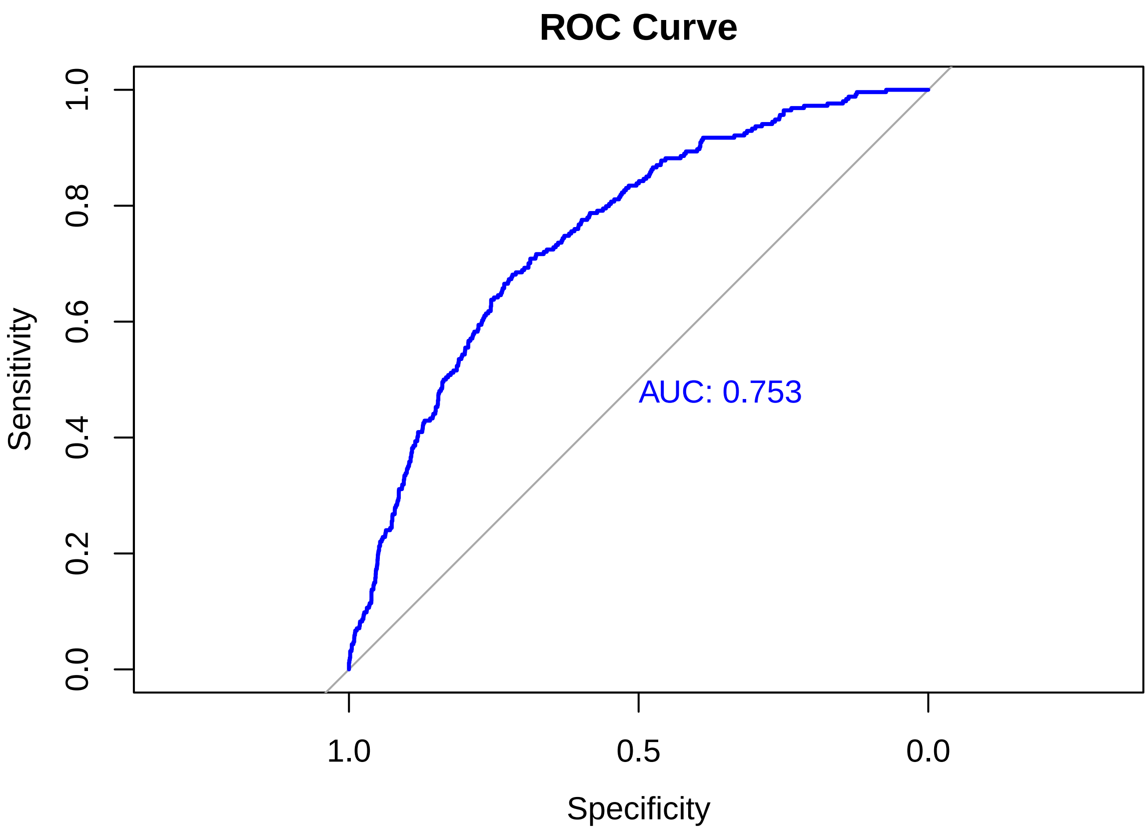
Task 4 - Discrimination

4a. Task: To discriminate between the cases and non-cases please plot the ROC curve and calculate the derived AUC including 95% confidence intervals.

Answer:

According to Dr. Thorgerdur Palsdottir presentation notes, the ROC Curve (Receiver operating curve) and and the AUC (Area Under the Curve) evaluate how well the predicted risk from our model distinguishes between patients with and without the disease.

```r
library(pROC)
#ROC curve object using pROC package
roc_curve <- roc(dc$chd69, dc$predicted_risk)

#Plotting ROC Curve
plot(roc_curve, col = "blue", main = "ROC Curve", print.auc = TRUE)
```




```
#AUC
auc_app <- auc(roc_curve)

# 95% CI for AUC
auc_ci <- ci.auc(roc_curve)
```

The derived AUC (area under the curve) is 0.753 which indicates good discrimination. The 95% Confidence Interval for AUC is $(0.7237, 0.7824)$. The ROC curve plots sensitivity (true positive rate) vs 1-specificity (false positive rate) across all possible thresholds. The diagonal gray line represents random selection (AUC = 0.5). Our curve is above this line and has a significant predictive value.

4b. Task: Find the threshold that maximizes the sum of the sensitivity and specificity. Please report the sensitivity and specificity at that threshold.

Answer: In general, we define sensitivity and specificity as:

$$\text{Sensitivity} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

$$\text{Specificity} = \frac{\text{True Negatives}}{\text{True Negatives} + \text{False Positives}}$$

Our code provides the following results:

```
     roc_curve.sensitivities roc_curve.specificities roc_curve.thresholds
2153               0.6811024                0.717851           0.09042859
```

The threshold that maximizes the sum of the sensitivity and specificity is 0.09042859, where sensitivity = 0.6811024 and specificity = 0.7178510. Practically this means that, at this threshold, the model correctly identifies 68.11% of the men who actually have CHD (Sensitivity) and correctly identifies 71.79% of the men who do not have CHD (Specificity).

At the optimal threshold of 9.04%, we have a good balance between the two metrics which are both around 70%.

4c. Task: AUC adjusted for optimism. To adjust for optimism in the predictions, we can use the bootstrapping method using 200 repetitions. Please calculate the adjusted AUC with 95% confidence intervals and compare it to the unadjusted AUC. Is there a difference in the unadjusted and the adjusted AUC?

Note: please do not use pre-programmed functions in R to answer this question.

Answer:

To adjust AUC for optimism, we used a bootstrap method with 200 repetitions, without pre-programmed functions in R. In our code, we started with bootstrap sampling, i.e. for each iteration we took a sample with replacement from the original dataset (boot_data). Then, we refit the model on each bootstrap sample and calculate the AUC on the bootstrap sample itself (boot_auc). We then applied the same fitted model to the original dataset and calculated the AUC there (test_auc).

The difference between these two AUCs (boot_auc – test_auc) is the optimism for that bootstrap sample and the adjusted AUC is the difference between the apparent AUC (auc_app) and the mean optimism (optimism_mean).

```r
#Set the seed for reproducability
set.seed("206")

optimism <- numeric(200)
for (i in 1:200){
  #Sample with replacement -> bootstrap sample boot_data
  boot_index <- sample(1:nrow(dc), replace = TRUE)
  boot_data <- dc[boot_index, ]

  #Fit model on boot_data
  boot_model <- glm( chd69 ~ age0 + cholmmol + sbp10 + bmi + smokerf +
                      arcus0 + dibpat0f + bmi:cholmmol,
                    data = boot_data,
                    family = binomial)


  #Find AUC for boot_data
  boot_data$boot_pred <- predict(boot_model,
                                 newdata = boot_data,
                                 type = "response")
  boot_roc <- roc(boot_data$chd69,boot_data$boot_pred)
  boot_auc <- auc(boot_roc)

  #AUC_test(b) on original data
  test_pred <- predict(boot_model,
                      newdata = dc,
                      type = "response")
  test_roc <- roc(dc$chd69,test_pred)
  test_auc <- auc(test_roc)
```

```r
  #Optimism(b) = AUC_boot(b) - AUC_test(b)
  optimism[i] <- boot_auc - test_auc
}

#Optimism_mean = average optimism(b)
optimism_mean <- mean(optimism)

#AUC_corrected = AUC_app - optimism_mean
auc_corrected <- auc_app - optimism_mean

#c(auc_app, auc_corrected)
```

Our code provides the following results:

Metric	AUC	95% CI
Unadjusted AUC	0.7530452	(0.7237, 0.7824)
Adjusted AUC	0.7461568	(0.7168, 0.7747)

As we can see here, the adjusted AUC is slightly less than the unadjusted AUC. This makes sense since we are subtracting the optimism in order to get the adjusted value. The confidence interval for the adjusted is shifted down as well.

The difference is relatively small and the overlapping confidence intervals do not imply a significant loss in model performance.

4d. Task: Cross validation. Use 10-fold cross-validation for a logistic model to obtain the adjusted AUC and compare to the unadjusted and adjusted AUC

Answer:

According to Dr. Thorgerdur Palsdottir, K-fold Cross-Validation (CV) is an internal validation technique used to estimate the model's expected prediction error. The 10-fold CV process starts with splitting the dataset (dc) into $K = 10$ folds. For each fold K, the model was trained on the remaining $K - 1$ folds. The trained model was used to predict probabilities on the held-out K-th fold. We then pooled all held-out predictions and computed the Cross-Validation Adjusted AUC (CV AUC) for all of them.

```r
folds <- sample(rep(1:10, length.out = nrow(dc)))

cv_pred <- rep(NA, nrow(dc))

for (i in 1:10) {

  train <- which(folds != i)
  test <- which(folds == i)

  # Train on K-1 folds
  cv_model <- glm(chd69 ~ age0 + cholmmol + sbp10 + bmi + smokerf + arcus0 + dibpat0f
                  + bmi:cholmmol,
                data = dc[train, ],
                family = binomial)

  cv_pred[test] <- predict(cv_model,
```

```
                                newdata = dc[test, ],
                                type = "response")
}

cv_roc <- roc(dc$chd69, cv_pred)
cv_auc <- auc(cv_roc)
```

In the following table, we compare the unadjusted AUC to the two internal validation methods:

Metric	Value	Difference from Unadjusted AUC
Unadjusted AUC	0.7530	–
Bootstrap Optimism Adjusted AUC	0.7462	-0.0068
Cross-Validation Adjusted AUC	0.7415	-0.0115

The Cross Validation adjusted AUC is less than the unadjusted AUC and is slightly less and the Bootstrap Optimism Adjusted AUC.

Both adjusted AUCs are lower than the unadjusted AUC which means there was indeed optimism in our initial predictions. However, the differences are relatively small which means that we can consider the model stable and not overfitted to the training data.

Task 5 - Calibration

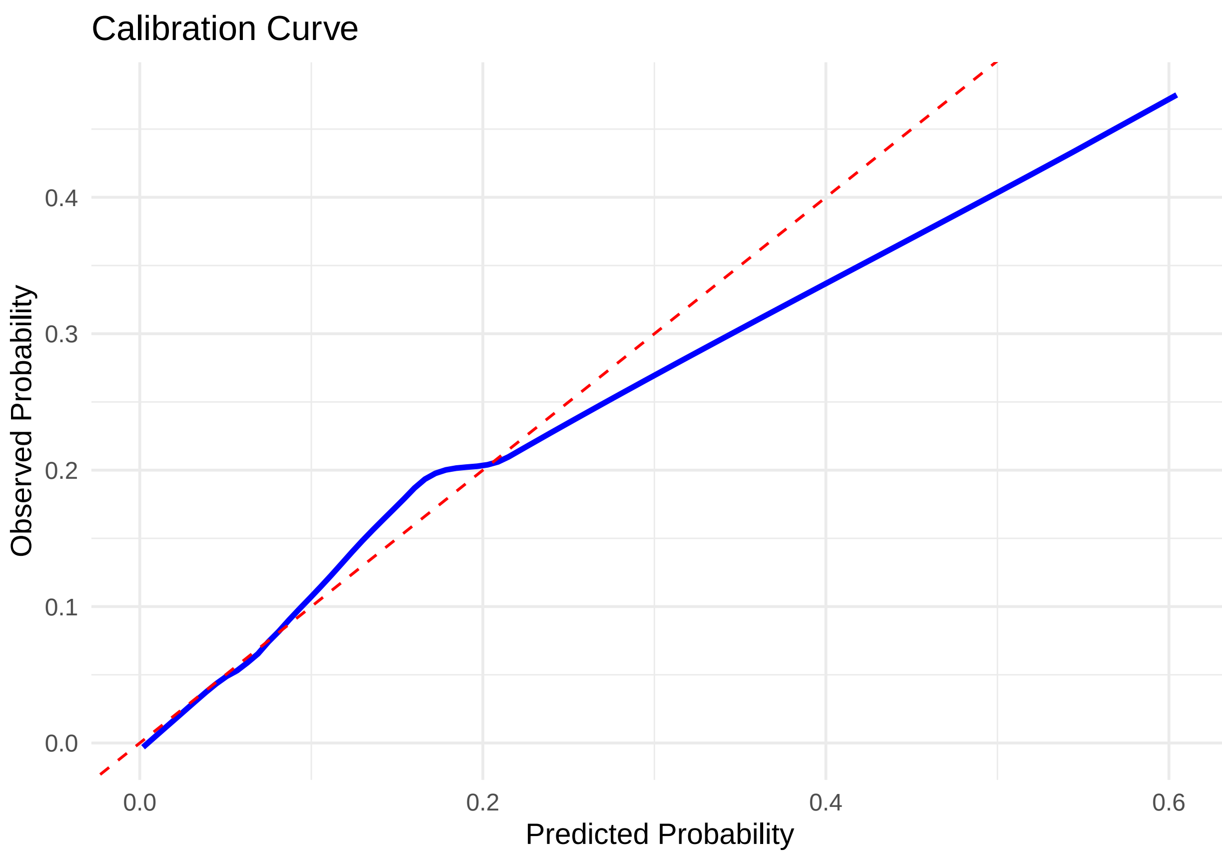
5a. Task: Calibration Curve. To evaluate the performance of the prediction model we have computed the AUC from ROC analysis, now please plot the calibration curve and report the slope and the intercept of the calibration curve. Use the model you chose before.

Answer:

According to Dr. Thorgerdur Palsdottir presentation notes, Calibration of a Prediction Model evaluates agreement between predicted risk and observed values in our data. We have three main tools for evaluation of calibration: the Calibration Plot, the Calibration-in-the-large and the Hosmer-Lemeshow test for goodness-of-fit

The calibration model is fitted by regressing the outcome (chd69) on the linear predictor (log-odds) from our main prediction model.

$$\text{logit}(P(\text{CHD}_i = 1)) = \text{Intercept} + \text{Slope} \cdot \text{logit}(\hat{P}_i)$$



The calibration model gives an Intercept value of 2.00×10^{-13} which is very close to 0 and means that there is no systematic bias (predictions are not systematically too high or too low). The Slope value which is approximately 1.00 also indicates an excellent fit.

The calibration curve shows how well the predicted risks from our model align with the observed risks of CHD in our cohort and compares predicted probabilities of CHD against observed outcomes, grouped into deciles. The diagonal line represents the perfect calibration. In the range between $0.0 - 0.20$ for the Predicted Probability, the black curve is very close to the ideal red one which indicates a good calibration for individuals predicted to have a low risk of CHD. However, as the predicted probability increases, our model's curve falls below the red line which is a sign of over-prediction, i.e. our model predicts risks that are higher than what is actually observed in the cohort. Our curve ends at a Predicted Probability of approximately 0.60. This means that our model is not predicting a risk higher than 60% because no one in our dataset was predicted to have a risk that high.

5b. Task: Please estimate the goodness of fit by the method of Hosmer and Lemeshow. Interpret the test results

Answer: According to Dr. Thorgerdur Palsdottir presentation notes, the Hosmer–Lemeshow (HL) works by grouping individuals by predicted risk, typically into 10 deciles and comparing the observed to the expected events within each group.

```
library(ResourceSelection)

# Hosmer-Lemeshow test on the final logistic regression model default: G=10 groups/deciles
hosmer_lemeshow_result<-hoslem.test(dc$chd69, fitted(interaction_model_bmi_cholmmol), g=10)
```

```r
# Print the result
print(hosmer_lemeshow_result)
```

```
    Hosmer and Lemeshow goodness of fit (GOF) test

data:  dc$chd69, fitted(interaction_model_bmi_cholmmol)
X-squared = 10.11, df = 8, p-value = 0.2574
```

In our case, the Hosmer-Lemeshow test provides a p-value equal to 0.2574 greater than the significance level 0.05. So, we fail to reject the null hypothesis (H_0) and we can conclude that there is no statistically significant difference between the observed event rates and the predicted event rates across the 10 risk groups. Hence, our logistic regression model has sufficient goodness of fit to the data. This finding is consistent with the intercept and slope values in Task 5a.

5c. Task: Create a prediction model only using variable agegroup as a predictor and estimate the discrimination

Answer:

The prediction model using only agegroup has the following form:

$$\text{logit}(P(\text{CHD}_i = 1)) = \beta_0 + \beta_1 \cdot \text{agegroup}_{[45,55)} + \beta_2 \cdot \text{agegroup}_{[55,60]}$$

The lowest age group, [39, 45) is the reference level.

```r
# prediction model using only agegroup
# data=dc complete case data
model_agegroup <- glm(chd69 ~ agegroup,data = dc,family = binomial)

# predicted probabilities
dc$predicted_risk_agegroup <- predict(model_agegroup,newdata = dc,type = "response")

library(pROC)

# ROC object
roc_agegroup <- roc(dc$chd69,dc$predicted_risk_agegroup, ci = TRUE )

auc_agegroup <- roc_agegroup$auc # AUC value

# Print the AUC value and the 95% CI
print(paste("AUC for agegroup model:", auc_agegroup))
```

```
[1] "AUC for agegroup model: 0.609068082260948"
```

```r
print(ci(roc_agegroup))
```

```
95% CI: 0.5765-0.6416 (DeLong)
```

The discrimination of this model measured by the AUC is 0.6091 which indicates that the model has poor discrimination, if we consider that a random guess has AUC = 0.50. The respective 95% confidence interval is $(0.5765, 0.6416)$

5d. Task: Please compare the discrimination to the model you used before with a statistical test and interpret the results.

Answer: We want to compare our final model which is:

$$\text{CHD} \sim \text{Age}+\text{Cholesterol}+\text{Systolic BP}+\text{BMI}+\text{Smoking Status}+\text{Corneal Arcus}+\text{Behavior Type}+\text{Cholesterol} : \text{BMI}$$

and we want to compare it to the simple age-only model which is

$$\text{CHD} \sim \text{AgeGroup}$$

We use the DeLong test for two correlated ROC curves:

```r
# DeLong test to compare the two AUCs
# order is important Compare the final model to the simpler model
delong_test_result <- roc.test(
  roc_curve,           # Model 1 - Our final Model
  roc_agegroup,        # Model 2 - Age-only Model
  method = "delong"
)

print(delong_test_result)
```

```

    DeLong's test for two correlated ROC curves

data:  roc_curve and roc_agegroup
Z = 8.3899, p-value < 2.2e-16
alternative hypothesis: true difference in AUC is not equal to 0
95 percent confidence interval:
 0.1103428 0.1776115
sample estimates:
AUC of roc1 AUC of roc2
  0.7530452   0.6090681
```

Our final model has a higher AUC of 0.7530 than the Age-Only Model's AUC which is 0.6091. Their difference (0.1439) indicates a much stronger predictive ability for our final model. Generally, the higher AUC, the better prediction the model has. Since the P-value is ($< 2.2 \times 10^{-16} < 0.05$), we definitely reject the null hypothesis that the two AUCs are equal. This indicates a statistically significant difference between the two AUCs. So, the additional traditional cardiovascular risk factors (cholesterol, SBP, BMI, smoking, arcus, and behaviour type) leads to a statistically better model compared to a model based only on the age group. We note that the AUCs are comparable since the predictions are made on same cohort.

```r
# --- Alternative/Supplementary Comparisons ---

pred_final <- predict(interaction_model_bmi_cholmmol , newdata = dc, type = "response")
pred_age   <- predict(model_agegroup,   newdata = dc, type = "response")

library(pROC)
roc_final <- roc_curve
roc_age   <- roc_agegroup
```

```r
# A. Likelihood Ratio Test
lrt_result <- anova(model_agegroup, interaction_model_bmi_cholmmol, test = "LRT")

# B. Confidence Interval Comparison
ci_final <- ci.auc(roc_final)
ci_age   <- ci.auc(roc_age)

print(lrt_result)
```

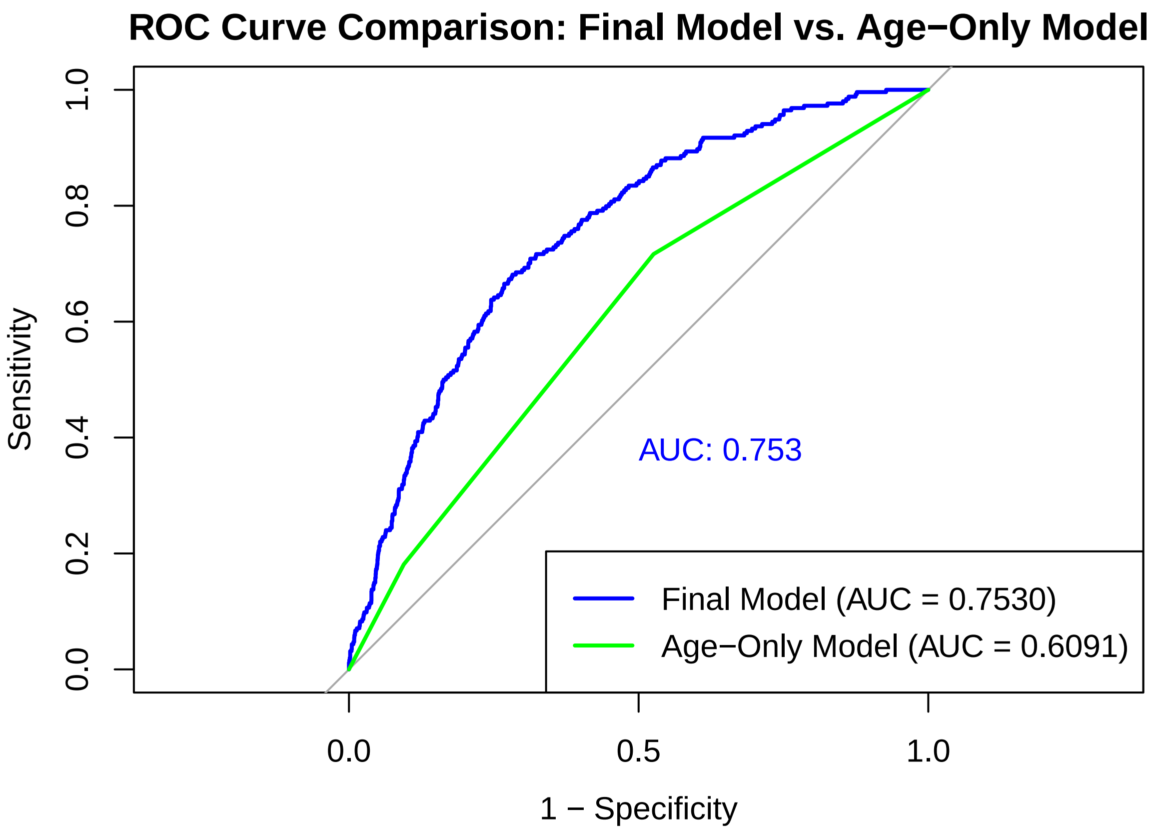
```
Analysis of Deviance Table

Model 1: chd69 ~ agegroup
Model 2: chd69 ~ age0 + cholmmol + sbp10 + bmi + smokerf + arcus0 + dibpat0f +
    bmi:cholmmol
  Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
1      3136     1723.3
2      3130     1570.3  6   153.01 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Additionally, the Likelihood Ratio Test (LRT) confirmed that the final model provides a significantly better fit to the data than the age-only model ($p < 0.001$), and the 95% confidence intervals of the AUCs do not overlap, as it can be verified from the following results:

Model	AUC 95% Confidence Interval
Final Model	(0.7237, 0.7824)
Age-Only Model	(0.5765, 0.6416)

5e. Task: Plot both ROC curves in one figure.

Answer:



From the combined plot, we see a large gap between the blue curve and the green curve which confirms that the additional traditional cardiovascular risk factors significantly ameliorate the model fit and our final selected model is obviously better than the simpler model.

Task 6 - Decision Curve Analysis

6a. Task: Plot the decision curve for the model you created and compare its net benefit to the default strategies provided in the function.

Answer:

According to Dr. Thorgerdur Palsdottir, in order to evaluate the clinical usefulness of a prediction model, we calculate the Net Benefit (NB):

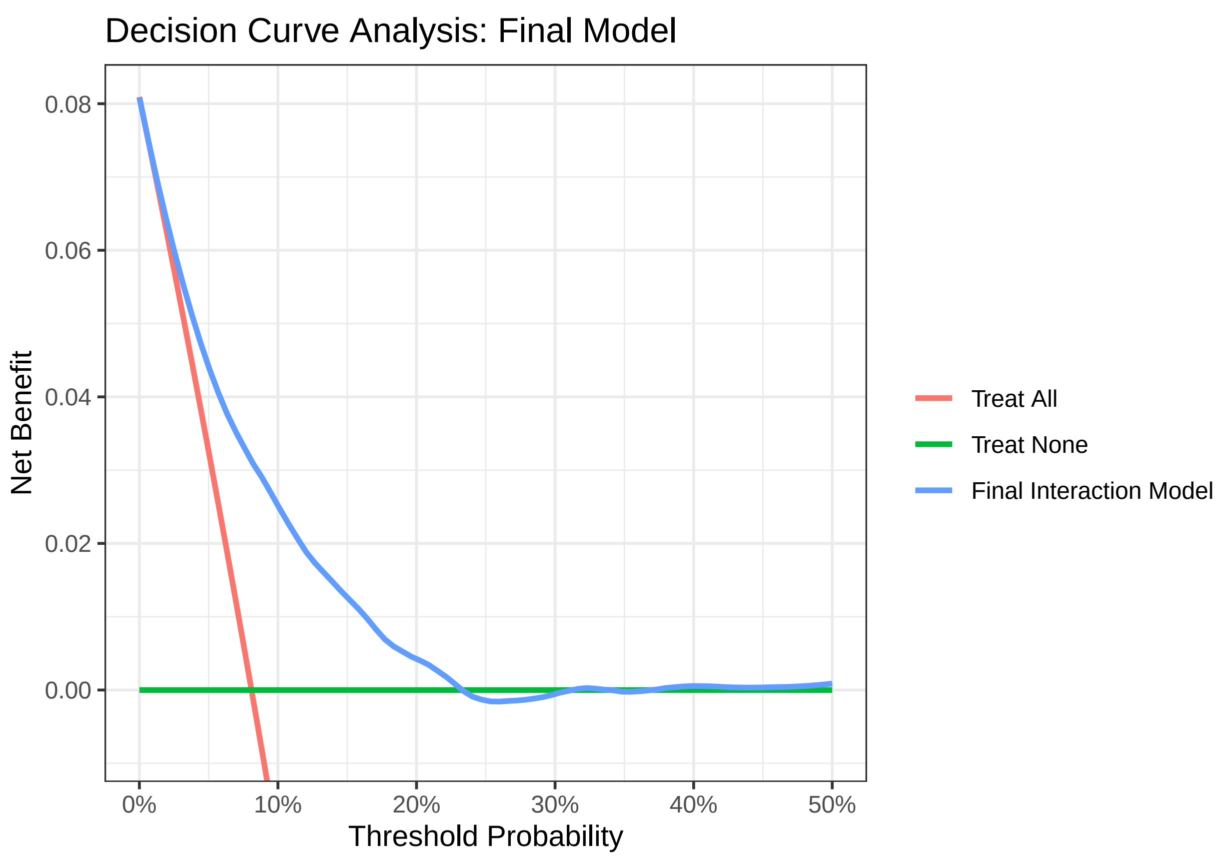
$$NB = \frac{TruePositives}{n} - \frac{FalsePositives}{n} \cdot \frac{p_t}{1 - p_t}$$

where n is the total number of patients and p_t the threshold probability.

We compute Net Benefit over a range of threshold probabilities p_t.

```
library(dcurves)

dca_result <- dca(chd69 ~ predicted_risk,
                  data = dc,
                  thresholds = seq(0, 0.50, by = 0.01),
                  label = list(predicted_risk = "Final Interaction Model"))
```

```r
plot(dca_result, smooth = TRUE) +
  ggplot2::labs(title = "Decision Curve Analysis: Final Model")
```



6b. Task: Is the model you have built clinically useful or clinically harmful? At what clinical thresholds is it beneficial?

Answer: At what clinical thresholds is it beneficial? The decision curve shows that the Final Interaction Model (Blue line) provides a higher Net Benefit compared to the default strategies of "Treat All" (Red) and "Treat None" (Green) across the threshold probability range of approximately 8% to 24%.

At low thresholds ($< 8\%$): The "Treat All" strategy performs similarly to the model, which is expected given the disease prevalence of 8%.

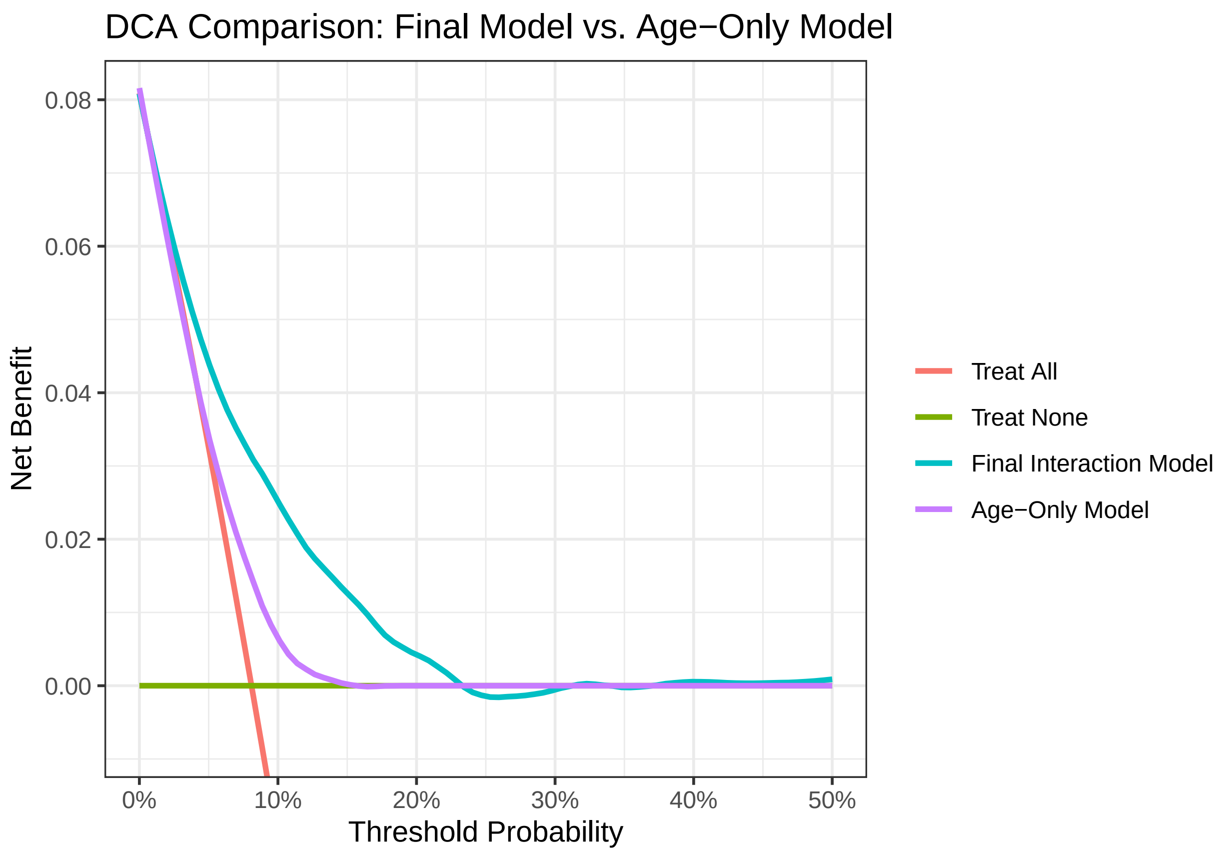
At clinically relevant thresholds (e.g., 10% - 20%): The model shows a clear positive Net Benefit, indicating its utility for screening decisions in this range.

At high thresholds ($> 25\%$): The Net Benefit drops slightly below zero. This corresponds with our findings in the Calibration Plot (Task 5a), where the model showed signs of over-prediction at higher probabilities. Therefore, the model should be used with caution if the decision threshold is set very high (above 25%), but it is robust for standard risk thresholds.

6c. Task: Differences in NB between models. Plot the model you created in 5.c and add it to the figure you have plotted of the net benefit in 6.a. Make comparisons between the two curves.

Answer: Here we compare the Net Benefit of our Final Model (with interactions) against the simpler Age-Only Model (from Task 5c).

```r
dca_comparison <- dca(chd69 ~ predicted_risk + predicted_risk_agegroup,
                      data = dc,
                      thresholds = seq(0, 0.50, by = 0.01),
                      label = list(predicted_risk = "Final Interaction Model",
                                   predicted_risk_agegroup = "Age-Only Model"))


plot(dca_comparison, smooth = TRUE) +
  ggplot2::labs(title = "DCA Comparison: Final Model vs. Age-Only Model")
```



Comparison with Age-Only Model: As shown in the comparison plot (Figure 2), the Final Interaction Model (Turquoise line) consistently demonstrates a higher Net Benefit than the Age-Only Model (Purple line).

While the Age-Only model's benefit drops to zero (becoming equivalent to "Treat None") at thresholds as low as 15%, the Final Model maintains a positive Net Benefit up to approximately 24%. This visualization confirms that incorporating traditional risk factors (Cholesterol, BMI, Smoking, etc.) significantly enhances the clinical utility of the prediction model compared to relying on age alone.

Task 7 - Discussion

7a. Task: Please discuss the clinical relevance of your model.

Answer: Our final prediction model demonstrates moderate to good clinical relevance based on the statistical metrics and decision curve analysis performed.

1. Discrimination: With an AUC of 0.753, the model has an acceptable ability to distinguish between individuals who will develop CHD and those who will not. This is better than the age-only model (AUC

0.61), confirming that including physiological and behavioral risk factors is essential for accurate risk stratification.
2. Decision Curve Analysis (DCA): The DCA is the strongest indicator of clinical utility. It shows that our model provides a positive Net Benefit compared to the default strategies ("Treat All" or "Treat None") across a probability threshold range of approximately 8% to 24%. This range is highly relevant because clinical intervention thresholds for cardiovascular prevention (e.g., initiating statin therapy or lifestyle interventions) often fall within this 10-20% risk range.
3. Actionability: The predictors included (Smoking, BMI, Cholesterol, SBP) are largely modifiable. This increases the clinical relevance, as the model not only predicts risk but highlights areas where intervention (e.g., smoking cessation, lowering blood pressure) could reduce that risk.

However, a limitation observed in the calibration plot is the tendency to over-predict risk at higher probabilities (>30%). This means the model might classify some high-risk patients as having a higher probability of CHD than they actually do, potentially leading to over-treatment in that specific subgroup.

7b. Task: What is needed for your model to be clinically relevant and useful in a clinical environment?

Answer: For this model to be deployed effectively in a real-world clinical setting, several requirements must be met:

1. External Validation: We have only performed internal validation (Bootstrap and Cross-Validation). Before clinical use, the model must be validated on an external, contemporary dataset. Our data is from 1960-1961 (WCGS); modern populations may have different baseline risks due to changes in diet, medication (e.g., widespread use of statins), and lifestyle.
2. Generalizability Check: The current dataset consists exclusively of middle-aged employed men. To be useful in a general clinical environment, the model needs to be tested and potentially updated to include women and a broader age range.
3. Integration into Workflow: To be useful, the model should be integrated into Electronic Health Records (EHR) as an automated calculator. Clinicians are unlikely to manually calculate the log-odds using the formula.
4. Clear Intervention Protocols: A risk score is only useful if it is paired with a decision guide. We need to define exactly what happens at different thresholds (e.g., "If Risk > 10%, start lifestyle counseling; if > 20%, consider medication").

7c. Task: What are the next possible steps for you as a researcher to ascertain that your model is clinically relevant?

Answer: To further ascertain the relevance and robustness of this model, the following steps should be taken:

1. Temporal Validation: Test the model on a more recent cohort (e.g., data from the 2000s or 2010s) to see if the regression coefficients hold true for modern patients, or if the model needs recalibration (updating the intercept and slope) to reflect current disease prevalence.
2. Subgroup Analysis: Investigate the model's performance across different ethnicities and socioeconomic groups, as the original WCGS data may lack diversity.
3. Impact Analysis: Conduct a study to see if using this model actually improves patient outcomes compared to current standard practice. Does showing the risk score to the doctor and patient actually result in behavior change or better prescribing patterns?
4. Updating Predictors: Consider investigating if adding modern biomarkers (e.g., hs-CRP) or genetic risk scores could significantly improve the AUC and Net Benefit beyond the current traditional risk factors.

References